ARYAN MANGAL

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EDUCATION

Carnegie Mellon University, Pittsburgh, USA — *May 2025*
*Master of Science in Robotics, **Advisor: Prof. Katia Sycara*** — *GPA 4.08/4*
Coursework: Computer Vision, Visual Learning and Recognition, Deep Learning Systems, Foundational Models.

Delhi Technological University, New Delhi, India — *May 2023*
Bachelor of Technology in Electronics And Communication Engineering — *GPA 9.5/10*
Fellowships: The J.N. Tata Endowment, DAAD WISE and MITACS Globalink Scholarships.

PUBLICATIONS

SymAnt: Neuro-Symbolic Diffusion Models for Action Anticipation — Workshop - **ICRA** 2025, **WACV** 2027
***Aryan**, Simon Stepputtis, Yaqi Xie, Hossein Nourkhiz Mahjoub, Katia Sycara* — Code Paper

AirLoc: Object-based Indoor Relocalization
***Aryan**, Bowen Li, Sebastian Scherer, Yun-Jou Lin, Chen Wang* — Code Paper

Grad-Lidar-SLAM: Fully Differentiable Global SLAM with Pose-Graph Optimization — Workshop - **IROS** 2022
***Aryan**, Dheeraj Vattikonda, Erqun Dong, Sabyasachi Sahoo* — Code Paper

EXPERIENCE

Machine Learning Engineer - Foundation Behavior Models — *July 2025 - Present*
Woven by Toyota, Palo Alto, CA

- Built **FBM (Foundation Behavior Model)**, a foundation model for driving behavior powering multiple downstream tasks via task-specific heads: distillation to the onboard behavior model, synthetic scene generation, and scene search.
- Built and deployed a neural simulation model trained with **offline RL** to enable inference-time control of agent behavior for closed-loop evaluation, and generated 2,000+ hours of synthetic scenario data.
- Accelerated **large-scale Transformer training** by migrating the compilation stack from TorchScript to TorchDynamo (torch.compile) and integrating FlashAttention-2 for memory-efficient attention.
- Built an agentic library that **automates org-wide ML workflows** (data generation, on-vehicle deployment validation, model training), cutting the ML model lifecycle from weeks to days and adopted by 700+ employees.
- Led integration of navigation intent into ML behavior model, enabling action decisions without rule-based heuristics.

Neuro-Symbolic Diffusion Models for Human Action Anticipation in HRI — *Aug 2023 - Present*
Advanced Agents Robotics Technology Lab, CMU - Research Assistant — Code Paper

- Built diffusion models to anticipate humans' long-term actions from short video, enabling proactive robot assistance.
- Achieved SOTA accuracy (+4-5%) at 80x smaller size and 3x faster training, enabling real-time robot deployment.

Research Intern - PyPose: A Library for Deep Learning with Physics-based Optimization — *Jan 2022 - Dec 2022*
The AirLab, CMU, Pittsburgh — Code CVPR Paper

- Developed a PyTorch library exposing physics-based optimization and Lie-group operations as differentiable layers.
- Built a batched differentiable IMU preintegration layer (SO(3) propagation, Jacobian covariance) for neural training.

PROJECTS

Rocky: Voice-driven manipulation and visual search with Spatial Memory — Code Demo

- Trained manipulation policies (ACT, GROOT) Can pickup task robust to arbitrary Can orientations on the ground.
- Designed an end-to-end voice-to-action system for multi-step pickup and visual search with spatial memory.

Xiao Hei: Vision-Language Navigation - IROS VLN Challenge — Code Website

- Built a VLN stack with Qwen-VL, YOLO-World+SAM perception for scene graphs, and frontier-based exploration.
- Achieved 80% classifiction accuracy on object-reference and spatial-memory questions.

Griffin: Autonomous Ground Vehicle (AGV) - Team Leader (3+ years) — Code

- Built ROS navigation/perception stack with SLAM, GPS-waypoint navigation, and LiDAR–IMU–GPS sensor fusion.
- Trained deep networks for lane following and obstacle avoidance; led the team to achieve 3rd position at IGVC 2023.

Needle: A Custom Autograd Library supporting Sparse Matrix Operations — Code

- Implemented an autograd library with custom CUDA kernels for sparse-dense matrix multiplications.
- Optimized Graph Neural Network training speed by 40% compared to dense matrix multiplications.

SKILLS

Languages & Frameworks: Python, C/C++, CUDA, PyTorch, JAX, Hugging Face Transformers, ROS/ROS 2
Robotics & Infra: VLAs, SLAM, Motion Planning, TensorRT, Triton, Distributed Training (DDP/FSDP), PEFT (LoRA)